MICROSTRIP PATCH ANTENNA FOR BLUETOOTH/ZIGBEE IoT NODES, FABRICATED ON FR-4 PCB

Technical Report — 2.45 GHz Microstrip Patch Antenna: Design, Simulation & Fabrication-Ready Validation in ANSYS HFSS

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Tool: ANSYS Electronics Desktop (HFSS) 2025 R2.4

1. INTRODUCTION & MOTIVATION

IoT sensor nodes built around Bluetooth Low Energy (BLE) and Zigbee (IEEE 802.15.4) need antennas that are compact, low-cost, and easy to mass-produce alongside the rest of the PCB assembly. A microstrip patch antenna printed directly onto standard FR-4 substrate meets all three needs: it shares the same low-cost material and fabrication process as the host PCB, requires no separate antenna component, and keeps the whole design low-profile and planar. The 2.4–2.4835 GHz ISM band is shared by BLE and Zigbee, so a single antenna tuned to 2.45 GHz can serve both radio standards — a common choice for smart-home, industrial, and wearable IoT nodes. This project designs, simulates, and validates such an antenna in ANSYS HFSS, with a layout intended for direct fabrication on 1.6 mm FR-4 PCB using a standard 50 Ω SMA edge connector.

2. PROJECT OBJECTIVES

- Design a compact 2.45 GHz inset-fed microstrip patch antenna on low-cost FR-4 substrate, impedance-matched to a 50 Ω feed line.
- Model and solve the structure in ANSYS HFSS using a Driven Modal solution across a 2.0–3.0 GHz discrete frequency sweep.
- Validate resonance, impedance matching (VSWR), and far-field radiation behaviour against ISM-band requirements for BLE and Zigbee.
- Finalize a parametric, manufacturable layout that is ready to be etched on FR-4 PCB with a standard SMA feed connector.

3. DESIGN PARAMETERS & SIMULATION METHODOLOGY

The antenna was modelled on a double-sided FR-4 epoxy substrate (εr = 4.4, tanδ = 0.02, h = 1.6 mm) with a full ground plane on the reverse layer, using a fully parametric layout (SubX, SubY, W, L, Wf, Fi, g) to permit rapid dimensional optimisation. Impedance matching between the 50 Ω microstrip line and the patch was achieved through an inset feed, and excitation was applied via a 50 Ω lumped port spanning the gap between the feed line and ground plane. The structure was enclosed in an air box with a radiation boundary condition to emulate open-space wave propagation. Key design values are summarised below.

Parameter	Symbol	Value	Purpose
Operating Frequency	f_o	**2.45 GHz**	Bluetooth 5.0 / BLE / Zigbee ISM band
Substrate	—	FR-4, εr 4.4	60 × 60 × 1.6 mm, low-cost PCB material
Patch Dimensions	W × L	37.41 × 28.85 mm	Sets input resistance and 2.45 GHz resonance
Inset Feed	Wf, Fi, g	3.02, 9.20, 1.00 mm	Matches 50 Ω microstrip line to the patch
Port / Boundary	—	Lumped, 50 Ω	Excitation with radiation-boundary air box; SMA at fabrication

4. RESULTS & PERFORMANCE

The simulated antenna resonates at 2.43 GHz — a 20 MHz shift from the 2.45 GHz design target, easily correctable by trimming the patch length (L) before fabrication — with a return loss of −16.40 dB and a −10 dB impedance bandwidth of 2.400–2.4835 GHz that fully covers the required ISM band. VSWR reaches 1.36 at resonance, indicating over 95% of incident power is delivered into the antenna. The far-field pattern shows clean broadside radiation with low cross-polarisation and back-lobe levels, and the complete HFSS solution passed all seven pre-solve validation checks (design settings, 3D model, boundaries/excitations, mesh, analysis setup, optimetrics, and radiation).

Metric	Simulated Result	Requirement	Status
Resonant Frequency	2.43 GHz (target 2.45 GHz)	2.400–2.4835 GHz	**Pass**
Return Loss (S_{11})	−16.40 dB at 2.43 GHz	≤ −10 dB	**Pass**
VSWR	1.36 at 2.43 GHz	≤ 1.5	**Pass**
Radiation Pattern	Broadside, low back-lobe	Peak at θ = 0°	**Pass**
HFSS Validation	7 / 7 categories passed	All green	**Pass**

5. CONCLUSION & FUTURE WORK

This project successfully delivers a validated, fabrication-ready 2.45 GHz microstrip patch antenna for Bluetooth- and Zigbee-class IoT nodes. The parametric layout is suitable for direct fabrication on a standard double-sided 1.6 mm FR-4 PCB with a 50 Ω SMA connector, making it ready to move from simulation to a physical, measurable prototype.

- Trim patch length (L) by a fraction of a mm to pull resonance from 2.43 GHz to exactly 2.45 GHz before fabrication.
- Fabricate the validated layout on FR-4 PCB and mount a 50 Ω SMA edge connector at the feed point.
- Measure S_{11}, VSWR, and the radiation pattern on a VNA and in an anechoic/open-field range, and compare against the simulated results.

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